The First Beat

Long before we understand what sound is, we feel and hear it together. In the womb, as our sensory systems develop, the fluid environment delivers the mechanical thud of our mother's heart and the physical impact of her steps directly through tissue and bone. That kinetic pulse is the very first rhythm any of us experience.

Thousands of years ago around early campfires, human beings used rhythm to connect with one another. It was not just a quiet sound carried through thin air; it was feet hitting packed earth, hands slapping wood and stone, and deep chest vibrations. Rhythm brought people together into a single tribe.

I was born deaf; lowercase "deaf" describes my audiological condition, while uppercase "Deaf" describes the linguistic and cultural community I call home. Growing up, I learned very quickly how our school system views sound. When music class started, the hearing kids stayed in the room together to sing, drum, and play bells. I was routinely pulled out of class and sent to the library to read or do worksheets by myself.

Hearing readers usually imagine the saddest possible scene here: an isolated child staring through a window at a joyful chorus. The truth is much funnier. At the time, I loved the library. My school interpreters only worked during lectures; they vanished during lunch and recess, so I quickly came to regard my hearing classmates as part of the background scenery, with roughly the same personal importance as trees. In the library, nobody demanded my attention on their flapping lips. I could hoard a stack of *Goosebumps* paperbacks in absolute peace.

That trade-off accelerated my reading so fast that by middle school I was bumped into high school honors literature classes. When I later transferred to the Oregon School for the Deaf, where my honors literature teachers were Deaf themselves, we could communicate directly and fluently. Toward my final years at OSD, my teacher and I were practically taking turns lecturing the class.

It was only much later that I started working out what that music room trade had really cost me. It was not the joy of singing, but the early, shared wiring that anchors you to other people. Today, I am anchored to hardware, logic analyzers, and my machines. If I had been born hearing, would that still be true? I do not know.

Music, in the emotional sense that hearing culture reveres, has always been an alien concept to me. I often imagine sitting in a packed concert hall while an orchestra plays something meant to move the room to tears. Within minutes, I would be bored, looking around at the crowd. The people in the rows beside me would have tears streaming down their cheeks. I would turn to a hearing friend next to me, see them crying too, and simply ask if they were okay.

For me, reaching an emotional state beyond neutral focus requires immense, deliberate friction: walking deep into the woods at twilight, watching the sunlight drop behind the canopy while the first stars surface, feeling the wind move across the grass. Hearing culture has an effortless shortcut. You press a button on a screen, inject specific acoustic frequencies straight into the ear canal, and trigger an instant chemical reaction. It looks almost like a digital drug; one to which I am completely immune.

For readers who grew up with music and watched it slip away as their hearing shifted, that silence is a profound, legitimate grief. To lose the songs that defined your youth or the voice of a loved one singing is an agonizing bereavement. For me, however, born profoundly deaf, music was never something I lost; it was something I never had. Intriguingly, neuroscientists recognize a condition called specific musical anhedonia, where individuals with perfectly intact hearing simply cannot experience emotional reward from music, finding it as neutral as static. For those mourning a lost acoustic world, haptics offers a physical way back in. For me, it offers an entirely new language.

The Limits of the Ear and the Strength of Touch

Hearing Health Foundation-funded researchers, including the flagship Hearing Restoration Project (HRP), are pursuing the ultimate frontiers in auditory science: regenerating sensory hair cells, investigating auditory neuron protocols, and targeting genetic causes of hearing loss. That research is vital, and every breakthrough brings medicine closer to biological cures.

Yet for someone like me, born deaf, those biological paths are not a simple fix. Because my brain developed without acoustic input, my auditory cortex underwent cross-modal neuroplasticity, reallocating its neural real estate to visual and tactile processing. For a prelingually deaf adult, attempting to decode the electrical stimulation of a late cochlear implant or an auditory brainstem implant requires years of exhausting auditory retraining. That path is a miraculous opportunity for infants and postlingually deaf adults, but for me, I have no desire to put my life on pause when I would much rather be in my workshop building machines.

For the millions of people who lose hearing gradually through age, noise exposure, or ototoxic medications, the biological ear fails unevenly. Long before total deafness sets in, the delicate hair cells that resolve high frequencies wear out. The crisp edge of a snare hit, the breath in a vocal, the metallic bite of a cymbal, and the sharp consonants that make speech intelligible are often the first elements to vanish. Hearing aids amplify what remains, but they cannot restore missing sensory cells, leaving acoustic sound compressed, blunt, and muddy. Cochlear implants do impressive work turning speech into electrical patterns, but a limited electrode array struggles to reproduce the dense harmonic layers of a full band or an orchestra.

Touch offers a second pathway that does not depend on the cochlea. Although human skin cannot follow a cymbal's shimmer up into the kilohertz, it does not need to. What fingertips excel at is timing, edge detection, and texture. Deep Pacinian corpuscles respond with exquisite

sensitivity to vibrations around 200 to 300 Hz, while shallow Meissner receptors pick up low-frequency flutter and rapid taps.

Routing sound through hardware, haptics functions like an acoustic crossover network. In high-fidelity speaker design, a crossover splits complex audio, sending low frequencies to a woofer and high frequencies to a tweeter. Haptics does the same thing across sensory modalities. Instead of forcing damaged hair cells to process smeared high-frequency transients, we hand off what touch does best: the exact attack of a drumstick, the sudden pluck of a bass string, how hard a note hits, and how fast it decays. A listener who can no longer hear the acoustic bite of a snare can feel the attack land crisply on a fingertip, perfectly aligned with the acoustic tones their hearing aids or cochlear implants provide.

This synergy between skin and ear is why our research team collaborates with Dr. Lina Reiss's Cochlear Implant and Hearing Loss Lab at Oregon Health & Science University (OHSU), exploring how residual acoustic hearing, cochlear implant stimulation, and tactile cues interact.

Tactile sound may also open up new avenues for hearing preservation; an intriguing possibility aligned with HHF's Keep Listening campaign. In theory, by delivering low-end punch and rhythmic impact directly to the body, haptic systems could let musicians, audio engineers, and concertgoers feel the physical power of bass and percussion without needing to crank acoustic volume to ear-damaging decibel levels. Whether tactile reinforcement actually leads listeners to turn down acoustic volume in real-world practice is an open question that researchers have yet to test, but the physical potential is compelling.

Biological cures and regenerative therapies promise to heal the future. Haptic engineering equips the present, providing an immediate physical bridge while laboratory science works toward tomorrow.

Nature and the Haptic Zoo

Treating touch as a secondary backup misses how communication works in nature. In biomusicology, researchers study how music first emerged in living things, tracing human rhythm back to the imitation of animal calls and physical movements.

By narrowing our cultural definition of music to airborne pressure waves collected by an ear canal, modern society has artificially severed sound from physical touch. In doing so, we have turned music into an acoustic abstraction, ignoring the evolutionary reality that rhythm began as a full-body seismic event. Omitting touch does not just exclude deaf and hard-of-hearing bodies; it strips music of its most primal, instinctive dimension.

Animals have always relied on physical vibration to communicate across space. Male alligators bellow with deep, low-frequency rumbles that dip directly into sub-audible infrasound, vibrating the surrounding water so intensely that the surface erupts into dancing droplets. Elephants

transmit and detect seismic rumbles through their footpads, communicating across miles of savannah.

This biological connection inspired an interactive installation I am building called the Haptic Petting Zoo. Instead of using acoustic speakers, the Petting Zoo uses custom mechanical vibrators and kinetic actuators. Visitors hold and touch physical animal forms that reproduce distinct tactile patterns from nature. You feel the heavy, low rumble of an alligator in your ribcage, or the rapid flutter of an insect wing directly on your fingertips.

To create these sensations, we use solenoids; electromagnetic actuators with an internal plunger that snaps forward and retracts, much like the mechanical flippers and pop-bumpers inside a classic pinball machine. When dialed in with precision, solenoids do not produce a cheap, buzzing vibration like a smartphone on a desk. They deliver sharp, percussive physical impacts with real physical mass.

When both deaf and hearing visitors try these devices, their expressions shift immediately. But there is a second thing that happens in the room, and it is the part I care about most: the moment raw mechanical vibration transforms into genuine emotion.

On its own, an isolated buzz is merely telemetry. But when a visitor pets a mechanical creature and solenoids tap out a rhythmic heartbeat beneath their palm, an ancient neurological switch flips. We are all primed by that first rhythm in the womb. If a tactile heartbeat accelerates into a rapid, frantic tempo, our nervous systems instinctively tighten with anxiety; if it slows to a steady, calm cadence, our shoulders drop and we relax, lulled toward calm. Because the vibration is delivered through a warm, living context, the brain stops analyzing the signal as cold machinery and begins feeling it as empathy. If we can attach emotional meaning to a kinetic pulse in an artificial beast, we can do the exact same thing with a kick drum, a bassline, or an ensemble groove.

Bringing Haptics to the Classroom

Bringing these ideas out of the workshop and into real classrooms led our team at CymaSpace's Universal Music Design program to an active research grant from the National Endowment for the Arts. Working alongside research director Dr. Shawn Trail and Dr. Lina Reiss's lab at OHSU, we are asking a direct question: How does a tactile, Deaf-centered STEAM music curriculum improve engagement, learning outcomes, and creative expression for K-12 students, particularly those who are deaf or hard-of-hearing?

Our starting point was GestoLumina, a modular interface we built at CymaSpace: fingertip rings equipped with micro-force sensors and vibration motors, paired with a wrist-mounted LED bracelet that translates audio into color. The first prototypes were laser-cut basswood, hot glue, and an ESP32 microcontroller whose pin layout was, frankly, evil.

In musical performance, timing is everything. While passive listeners can tolerate wider sensory discrepancies, active musicians notice timing delays at roughly fifteen to twenty milliseconds, where rhythmic synchronization starts to degrade. That tight timing budget leaves very little margin. On our test bench, our firmware and driver circuitry process gestures and fire trigger pulses in under 2.3 milliseconds. Mechanical actuator rise time adds additional latency on top of that depending on the motor design, which is why characterizing end-to-end physical onset with bench accelerometers is part of our ongoing work. Keeping electrical latency tiny ensures the physical pulse has the best possible chance to align with the visual flash and acoustic sound.

The moment I knew the concept worked was during a duet. My musical partner is a cochlear implant user, while I am completely deaf. We did not need to speak, and we did not need to hear each other to stay in sync. I squeezed a rhythm between my thumb and forefinger, and my partner felt that exact kinetic impact land on their finger; multitrack monitoring conducted entirely through skin. Then they tapped a syncopated polyrhythm back onto mine.

Now, we are taking these tools into public school classrooms with our regional inclusive education partners, including Columbia Regional Inclusive Services (CRIS). Students explore musical scales, dynamic variations, and complex rhythms both with and without haptic feedback, while we track their engagement through structured surveys and observations. We are not simply handing deaf and hard-of-hearing kids a novel gadget; we are testing whether tactile physics keeps them anchored in the room alongside their hearing peers.

True Sonic Agency

Throughout our research, we focus heavily on the concept of sonic agency: the power to participate in the world of sound on your own terms. For generations, traditional music culture told deaf and hard-of-hearing people that sound did not belong to them. If you could not hear acoustic pitch through an ear canal, you were excluded from the conversation. We disagree with that completely.

Sound is physical energy, but music is the deliberate architecture of that energy. Music is not merely a sequence of acoustic pitches; it is the physical relationship of attack, duration, syncopation, tension, and decay over time. Much of the pleasure human beings derive from music comes from pattern recognition and musical expectancy: our brains delight in anticipating exactly when a note will land, predicting the pocket of a groove, and feeling the release of a delayed beat.

When a drummer strikes a snare or a synthesist modulates a low-frequency oscillator, they are orchestrating physical shockwaves. When properly engineered hardware delivers those impulses directly to the skin with very low latency, a student is not receiving a pale acoustic imitation; they are experiencing music's structural foundation. They can feel the pocket, track the groove, and shape the dynamics themselves.

Why, then, does hearing culture experience music with such overwhelming emotion, while someone like me remains neutral?

Part of the answer, I suspect, brings together neurodevelopment and classical conditioning. In early childhood, our brains rely on critical developmental windows to build the cross-modal scaffolding that binds our senses together. When that early sensory access is in place, life experiences take over through classical conditioning, particularly in its evaluative form: pairing a neutral stimulus with an emotional event until the signal itself evokes the feeling. It is the psychological mechanism famously demonstrated by Ivan Pavlov, who paired the sound of a bell with food until the chime alone triggered a physiological reaction.

Nobody is born weeping at the sound of a cello. A hearing person cries at a cello because that melody was playing at a funeral, during a wedding, or in the car on a late drive home with someone they loved. Hearing culture spends a lifetime pairing acoustic frequencies with joy, heartbreak, grief, and camaraderie. Sound is present during all of life's most profound moments, repeated hundreds of times until the acoustic frequency alone can ring the emotional bell. I had rich, meaningful experiences throughout my life, too; but because I was isolated from sound, the acoustic frequency was never paired with the emotion. I simply never heard the bell.

Rhythm possesses an ancient, hardwired pull, but what was taken from me when I was sent to the library was not the musical scale. It was the room: the shared, synchronized, full-body experience where sound is forged into human belonging.

That is the part we can give back. I do not build these instruments because I have fallen in love with melody, or because I feel a poetic calling to teach deaf children how to sing. I build them because access should never be dictated by the limits of a biological sensor. When we put low-latency haptic instruments into the hands of deaf and hard-of-hearing students alongside their classmates, we give that sensory scaffolding and emotional connection a chance to take root early. What that rhythm ends up meaning is entirely theirs to decide: whether that means developing an emotional language I never had, or simply possessing the agency to stand in the room, fully equipped, and make up their own minds.